
Online Examination Report

Date:	25.07.2014
Attempts:	1
Examinee:	Menya Komyhendo, Ruth
Date of birth:	11/28/1988
Subject:	080- Principles of flight
Result:	37,5 %
Time used:	00:19:09 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	226	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
2	211	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
3	130	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
4	209	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
5	190	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
6	233	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
7	240	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
8	232	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
9	210	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
10	147	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
11	204	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
12	205	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
13	168	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
14	194	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
15	113	81.1 SUBSONIC AERODYNAMICS		0,00	1,00
16	82	81.1.1.1 Laws and definitions		1,00	1,00
17	17	81.1.1.5 Wing shape		0,00	1,00
18	30	81.1.2 Two-dimensional airflow around an aerofoil		0,00	1,00
19	100	81.1.3.1 The lift coefficient C_l		0,00	1,00
20	47	81.1.4.1 Streamline pattern		0,00	1,00
21	93	81.1.4.2 Induced drag		0,00	1,00
22	90	81.1.4.2 Induced drag		0,00	1,00
23	74	81.1.4.2 Induced drag		1,00	1,00
24	34	81.1.4.2 Induced drag		1,00	1,00
25	38	81.1.8.2 The stall speed		0,00	1,00
26	91	81.2.1.3 Influence of temp and alt on M		1,00	1,00
27	39	81.2.3.4 Effect on pitching moment		0,00	1,00
28	102	81.4 STABILITY		0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	14	81.4.1.1	Basics and definitions	0,00	1,00
30	87	81.4.3.3	Neutral point	0,00	1,00
31	63	81.4.6.3	Dutch roll	1,00	1,00
32	99	81.4.6.3	Dutch roll	1,00	1,00
33	72	81.5.4.5	Means to avoid adverse yaw:	1,00	1,00
34	6	81.6.1.1	Flutter	0,00	1,00
35	76	81.7.1.1	Relevant propeller parameters	0,00	1,00
36	8	81.7.4.2	Gyroscopic precession	0,00	1,00
37	33	81.7.4.2	Gyroscopic precession	0,00	1,00
38	89	81.8.1.3	Straight steady descent	0,00	1,00
39	19	81.8.1.4	Straight steady glide	1,00	1,00
40	62	81.8.2.3	Forces parallel to the lateral axis	1,00	1,00
Total:38%				15,00	40,00

Attachment 2: List of result by topic

Licence

Date: 25.07.2014

Examinee: Menya Komyhendo, Ruth

Location: Uganda Civil Aviation Authority

Your Result: 15,00 from 40,00 Points (37,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	15,00	40,00	37,50
81.1. SUBSONIC AERODYNAMICS	11	3,00	11,00	27,27
81.2. HIGH SPEED AERODYNAMICS	2	1,00	2,00	50,00
81.4. STABILITY	5	2,00	5,00	40,00
81.5. CONTROL	1	1,00	1,00	100,00
81.6. LIMITATIONS	1	0,00	1,00	0,00
81.7. PROPELLERS	3	0,00	3,00	0,00
81.8. FLIGHT MECHANICS	3	2,00	3,00	66,67
Total	40	15,00	40,00	37,50

Candidate: Ruth Menya Komyhendo
Date of birth: 1988-11-28

Subject: 080- Principles of flight
Date: 2014-07-25

1

During a wing stall, the wing tips of a weight shift aircraft are

- ☐ ineffective for stall recovery.
- ☐ effective only when combined with maximum engine output.
- ☒ effective for stall recovery.

2

If swirling dust, leaves, or debris indicate a strong thermal on the final approach to a landing, it is recommended that the glider pilot

- ☐ open the spoilers and reduce the airspeed.
- ☒ open the spoilers and maintain a constant airspeed.
- ☐ close the spoilers and increase the airspeed.

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3

What type of stability does the horizontal stabilizer provide during flight?

☐ Airspeed.

☐ Lateral.

☒ Longitudinal.

4

Unless adequate speed control is maintained during the turn to base and the final approach for a landing into the wind, which would most likely occur if a steep wind gradient existed?

- ☒ The airspeed on final approach would increase, causing the glider to overshoot the desired landing spot.
- ☐ The desired landing spot would be undershot or the glider would stall.
- ☐ The wingtip on the outside of the turn would stall before the wingtip on the inside of the turn.

5

Maximum gliding distance of an aircraft is obtained when

- ☐ parasite drag is the least.
- ☒ induced drag equals the coefficient of lift.
- ☐ induced drag and parasite drag are equal.

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6

After being handed off to the final approach controller during a 'no-gyro' surveillance or precision approach, the pilot should make all turns

☐ standard rate.

☒ based upon the groundspeed of the aircraft.

☐ one-half standard rate.

7

(Refer to Figure 146.) Identify the system that has failed and determine a corrective action to return the airplane to straight-and-level flight.

- ☐ Vacuum system has failed; reduce power, roll left to level wings, and pitchup to reduce airspeed.
- ☒ Static/pitot system is blocked; lower the nose and level the wings to level-flight attitude by use of attitude indicator.
- ☐ Electrical system has failed; reduce power, roll left to level wings, and raise the nose to reduce airspeed.

See Attachments:

1. instrument_146.jpg

figure 146

(Attachment
No.1)

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8

During a 'no-gyro' approach and prior to being handed off to the final approach controller, the pilot should make all turns

☒ **one-half standard rate unless otherwise advised.**

☐ **standard rate unless otherwise advised.**

☐ **any rate not exceeding a 30 degrees bank.**

9

A rule of thumb for flying a final approach is to maintain a speed that is

- ☐ twice the glider's stall speed plus half the estimated windspeed.
- ☐ 50 percent above the glider's stall speed plus half the estimated windspeed.
- ☒ 50 percent above the glider's stall speed, regardless of windspeed.

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10

When a slight upward or negative flap deflection is used, the result is

☐ increased drag.

☐ decreased lift.

☒ decreased drag.

11

How can excessive towline slack that is allowed to develop during a glider tow be eliminated?

- ☐ Execute a shallow banked coordinated turn to either side.
- ☐ Increase pitch attitude until towline becomes taut.
- ☒ Yaw the nose to one side with rudder while keeping the wings level with ailerons.

12

What could result if a glider pilot releases while in the low-tow position during an aerotow?

- ☐ Tow ring may strike and damage the glider after release.
- ☐ Glider may be forced into the towplane's wake turbulence.
- ☒ Nose of the glider would tend to pitch up after release.

13

Which statement is true regarding propeller efficiency? Propeller efficiency is the

- ☐ ratio of thrust horsepower to brake horsepower.
- ☒ difference between the geometric pitch of the propeller and its effective pitch.
- ☐ actual distance a propeller advances in one revolution.

14

If the CG of an aircraft is moved from the aft limit to beyond the forward limit, how will it affect the cruising and stalling speed?

- ☐ Decrease both the cruising speed and stalling speed.
- ☐ Increase both the cruising speed and stalling speed.
- ☒ Decrease the cruising speed and increase the stalling speed.

15

When the angle of attack of a symmetrical airfoil is increased, the center of pressure will

- ☒ have very little movement.
- ☐ remain unaffected.
- ☒ move aft along the airfoil surface.

16

The SI units of air density (I) and force (II) are..... &... respectively:

☐ (I) kg / m^2 , (II) kg .

☐ (I) N / kg , (II) kg .

☒ (I) kg / m^3 , (II) N .

☐ (I) N / m^3 , (II) N .

17

Taper ratio of a wing is the ratio between:

- ☐ root chord and tip chord.
- ☒ wing span squared and wing area.
- ☐ tip chord and root chord.
- ☐ mean geometric chord and wing span.

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18

(Please view annex 081-0005 issue date July 2004)

How are the speeds at point 1 and point 2 related in the figure to the relative wind/airflow V?

☒ $V1 < V2$ and $V2 < V$.

☐ $V1 > V2$ and $V2 < V$.

☐ $V1 = 0$ and $V2 = V$.

☐ $V1 = 0$ and $V2 > V$.

See Attachments:

1. 081-0005.gif

Annex

(Attachment
No.2)

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19

(Please view annex 081-0006 issue date July 2004)

The point in the annex corresponding to CL for minimum horizontal flight speed is:

☐ point a.

☐ point d.

☒ point b.

☐ point c.

See Attachments:

1. 081-0006.gif

Annex

(Attachment
No.3)

20

Which of these statements about the strength of wing tip vortices are correct or incorrect?

- I. Assuming no flow separation, the strength of wing tip vortices decreases as the angle of attack decreases.
- II. The strength of wing tip vortices increases as the aspect ratio decreases.

- ☒ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is correct.

21

Which of these statements about induced drag are correct or incorrect?

- I. An elliptical spanwise lift distribution generates more induced drag than a rectangular lift distribution.
- II. Induced drag decreases with decreasing aspect ratio.

- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☒ I is correct, II is incorrect.

22

Excluding constants, the coefficient of induced drag (C_{Di}) is the ratio of:

- ☒ C_L and C_D .
- ☐ C_L^2 and S (wing surface).
- ☐ C_L and b (wing span).
- ☐ C_L^2 and AR (aspect ratio).

23

An aeroplane transitions from steady straight and level flight into a horizontal co-ordinated turn with a load factor of 2, the speed remains constant and the:

- ☒ induced drag increases by a factor of 4.
- ☐ lift increases by a factor of 4.
- ☐ total drag increases by a factor of 4.
- ☐ angle of attack increases by a factor of 1/4.

24

Which of these statements about induced drag are correct or incorrect?

- I. An elliptical spanwise lift distribution generates less induced drag than a rectangular lift distribution.
- II. Induced drag decreases with decreasing aspect ratio.

☒ I is correct, II is incorrect.

☐ I is incorrect, II is correct.

☐ I is incorrect, II is incorrect.

☐ I is correct, II is correct.

25

The stall speed increases, when: (all other factors of importance being constant)

☐ minor altitude changes occur e.g. 0-10.000 ft.

☐ weight decreases.

☐ pulling out of a dive.

☒ spoilers are retracted.

26

If the altitude is increased and the TAS remains constant in the troposphere under standard atmospheric conditions, the Mach number will:

☐ increase or decrease, depending on the type of aeroplane.

☒ increase.

☐ not change.

☐ decrease.

27

Which of these statements about "tuck under" are correct or incorrect?

- I. A contributing factor to "tuck under" is a forward movement of the centre of pressure of the wing.
- II. A contributing factor to "tuck under" is an increase in the downwash angle at the location of the horizontal stabiliser.

☐ I is incorrect, II is incorrect.

☒ I is incorrect, II is correct.

☐ I is correct, II is correct.

☐ I is correct, II is incorrect.

28

The capability of an aircraft to respond to a pilot's inputs, especially with regard to flightpath and attitude, is

☐ response.

☒ maneuverability.

☐ controllability.

29

A statically unstable aeroplane:

- ☒ can never be dynamically stable.
- ☐ will become stable at high speed.
- ☒ reduces its instability as the CG moves aft.
- ☐ is always dynamically stable.

30

Which of the following statements about static longitudinal stability is correct?

- I. A requirement for positive static longitudinal stability of an aeroplane is, that the neutral point is behind the centre of gravity.
- II. A wing with positive camber provides a positive contribution to static longitudinal stability, when the centre of gravity of the aeroplane is in front of the aerodynamic centre of the wing.

☐ I is incorrect, II is correct.

☐ I is incorrect, II is incorrect.

☐ I is correct, II is correct.

☒ I is correct, II is incorrect.

31

Which aeroplane behaviour will be corrected by a yaw damper?

☐ Spiral dive.

☐ Flutter.

☒ Dutch roll.

☐ Buffeting.

32

An aeroplane is sensitive to Dutch roll when:

- ☒ **static lateral stability is much more pronounced than static directional stability.**
- ☐ **an aeroplane has anhedral.**
- ☐ **static lateral stability is much less pronounced than static directional stability.**
- ☐ **static stability is less pronounced than dynamic stability.**

33

An example of differential aileron deflection during initiation of left turn is:

- ☐ Left aileron: 2° up.
Right aileron: 5° down.
- ☐ Left aileron: 2° down.
Right aileron: 5° up.
- ☒ Left aileron: 5° up.
Right aileron: 2° down.
- ☐ Left aileron: 5° down.
Right aileron: 2° up.

34

Which of these statements about flutter are correct or incorrect?

I. If flutter occurs, IAS should be reduced.

II. Resistance to flutter increases with increasing wing stiffness.

☐ I is correct, II is correct.

☒ I is incorrect, II is incorrect.

☐ I is incorrect, II is correct.

☐ I is correct, II is incorrect.

35

The difference between a propeller's blade angle and its angle of attack is called:

☐ the propeller angle.

☐ the helix angle.

☒ propeller slip.

☐ the effective pitch.

36

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

- I. Pitch up produces left yaw.
II. Right yaw produces pitch up.

☐ I is incorrect, II is correct.

☐ I is correct, II is incorrect.

☐ I is incorrect, II is incorrect.

☒ I is correct, II is correct.

37

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

- I. Pitch up produces right yaw.
II. Right yaw produces pitch down.

☐ I is incorrect, II is correct.

☒ I is correct, II is incorrect.

☐ I is correct, II is correct.

☐ I is incorrect, II is incorrect.

38

During a straight steady descent, lift is:

- ☐ less than weight,
- ☒ greater than weight because it must additionally balance a component of drag.
- ☐ equal to weight to maintain a constant speed.
- ☐ less than weight

39

The descent angle of a given aeroplane in a steady wings level glide has a fixed value for a certain combination of:

☒ configuration and angle of attack.

☐ altitude and configuration.

☐ configuration and mass.

☐ mass and altitude.

40

VMCA is certified with a bank angle of not more than 5° towards the operating engine (live engine low) because:

- ☐ at 5° bank (live engine low), sideslip is zero.
- ☐ the slip indicator at 5° bank (live engine low) is centred.
- ☒ although more bank reduces VMCA, too much bank may lead to fin stall.
- ☐ more than 5° bank (live engine low) would not reduce VMCA.

Attachment No. 1

Questions 7

figure 146

instrument_146.jpg

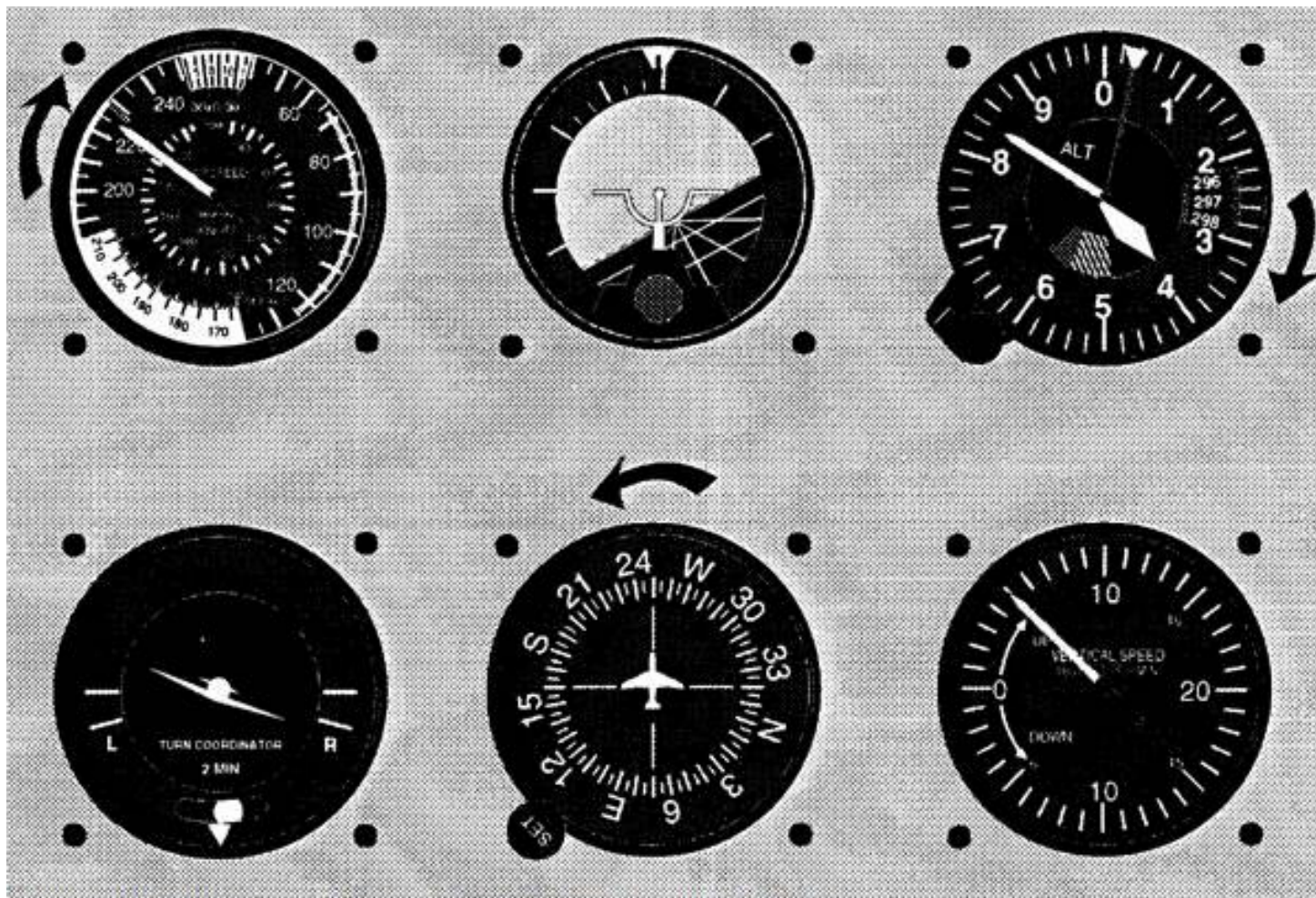


Figure 146. Instrument Sequence (System Failed) © ASA

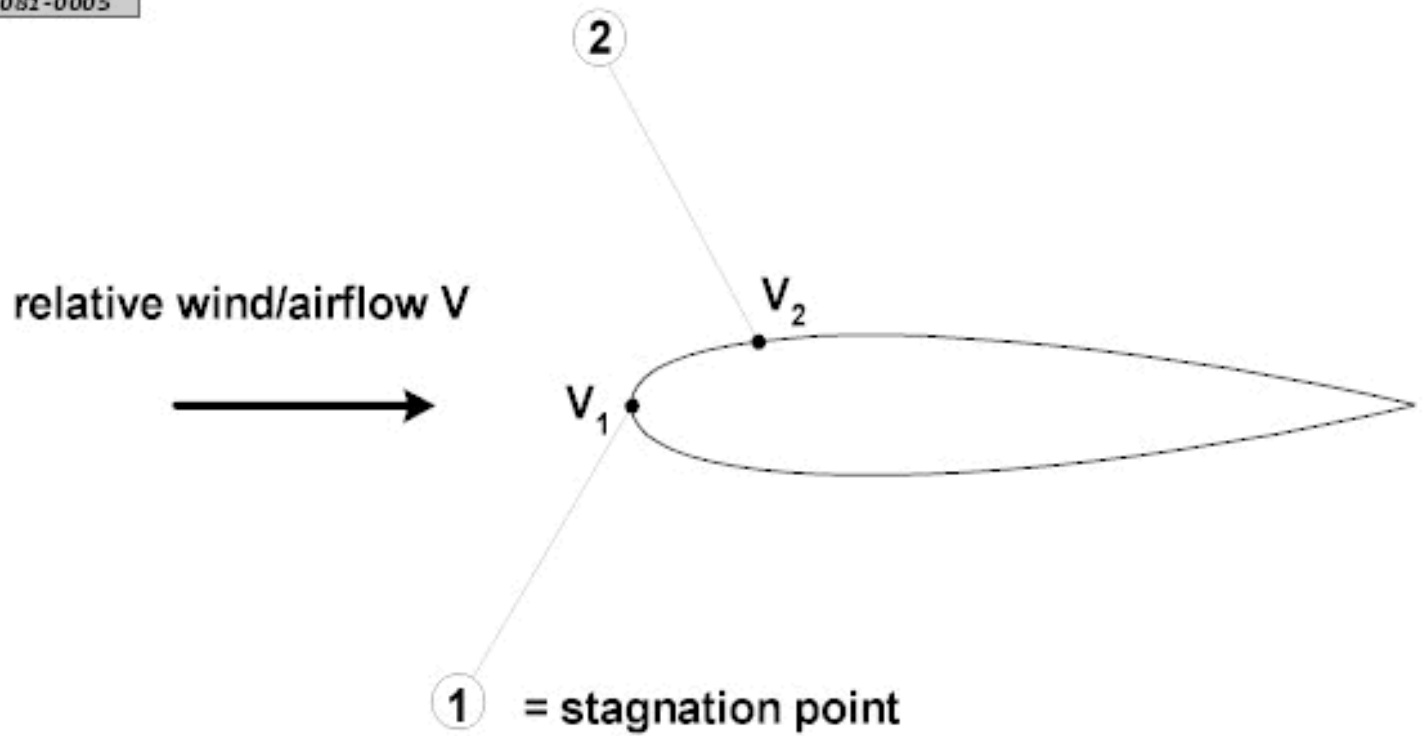
Attachment No. 2

Questions 18

Annex

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Attachment No. 3

Questions 19

Annex

081-0006.gif

081 - 0006

